# When Large Vision-Language Model Meets Large Remote Sensing Imagery: Coarse-to-Fine Text-Guided Token Pruning

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https://github.com/VisionXLab/LRS-VQA

2025 IEEE/CVF International Conference on Computer Vision (ICCV) | 979-8-3315-8775-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCV51701.2025.00860

## Abstract

*Efficient vision-language understanding of large Remote Sensing Images (RSIs) is meaningful but challenging. Current Large Vision-Language Models (LVLMs) typically employ limited pre-defined grids to process images, leading to information loss when handling gigapixel RSIs. Conversely, using unlimited grids significantly increases computational costs. To preserve image details while reducing computational complexity, we propose a text-guided token pruning method with Dynamic Image Pyramid (DIP) integration. Our method introduces: (i) a Region Focus Module (RFM) that leverages text-aware region localization capability to identify critical vision tokens, and (ii) a coarse-to-fine image tile selection and vision token pruning strategy based on DIP, which is guided by RFM outputs and avoids directly processing the entire large imagery. Additionally, existing benchmarks for evaluating LVLMs' perception ability on large RSI suffer from limited question diversity and constrained image sizes. We construct a new benchmark named LRS-VQA, which contains 7,333 QA pairs across 8 categories, with image length up to 27,328 pixels. Our method outperforms existing high-resolution strategies on four datasets using the same data. Moreover, compared to existing token reduction methods, our approach demonstrates higher efficiency under high-resolution settings.*

## 1. Introduction

Benefiting from the rapid advancement of Large Language Models (LLMs) [2, 11, 14, 17, 75], Large Visual-Language Models (LVLMs) have demonstrated strong capabilities in perceiving and understanding visual information in text-based multimodal interactions [9, 29, 41, 65]. Currently, LVLMs have been extensively studied and applied across various fields like remote sensing (RS) intelligent interpretation [12, 32, 53, 73, 87, 94].

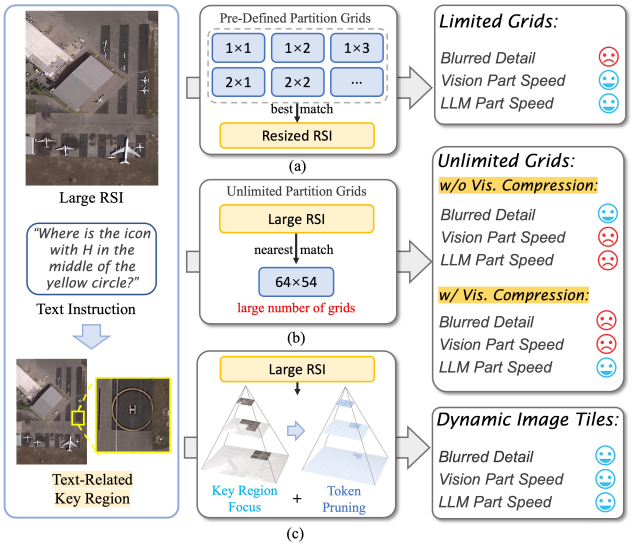



Figure 1. High-resolution strategy comparison for modular LVLMs. (a) and (b) show that existing grid-based cropping methods face challenges when processing large RSIs. (c) The proposed dynamic pyramid-based token pruning strategy can dynamically select image tiles of key regions related to the input text, balancing image detail and computational cost.

Advances in satellite imaging technology allow for the acquisition of large remote sensing images (RSIs) that cover extensive ground areas and contain detailed land use information [18, 38]. Therefore, language-based intelligent analysis of large RSIs is valuable for applications such as complex scene understanding [37, 50], urban planning [35, 64], infrastructure development [70], and monitoring [93].

Some RS LVLMs handle high-resolution input through position embedding interpolation [27, 86]. Additionally, various dynamic high-resolution strategies designed for general LVLMs [33, 41, 80] can be adapted to RS LVLMs, such as EarthDial [61] (based on InternVL1.5 [10])

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DOI 10.1109/ICCV51701.2025.00860

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and GeoPixel[57] (using InternLM-XComposer2.5 [84]). However, these approaches for high-resolution image processing are limited when handling large RSIs exceeding 10,000×10,000 pixels. As depicted in Fig. 1, these methods typically employ limited pre-defined grid partitions, leading to detail loss when large RSIs are excessively downsampled. Conversely, unlimited grids suffer from prohibitive time and memory costs (e.g., over 5 million vision tokens for a large RSI in LLaVA-1.5 [41]). This necessitates balancing resolution and computational efficiency when processing large RSI.

To overcome the above limitations of existing strategies in handling large RSIs, we propose a text-guided token pruning strategy that consists of two key components: a Region Focus Module (RFM) and a Dynamic Image Pyramid (DIP). The RFM identifies text-relevant key vision tokens by leveraging the capability distilled from the LLM component. Based on the RFM's output, we select critical image tiles and conduct token pruning in a coarse-to-fine manner within the DIP. This enables the LVLM to focus and zoom in key areas, avoiding to process all visual tokens, thereby facilitating flexible and efficient inference.

Moreover, benchmarks for LVLMs in understanding large RSIs remain insufficiently developed. Although MME-Realworld [91] includes large RSIs with high-quality manually annotated questions, its RS subset suffers from limited question diversity and image sizes. To more comprehensively characterize the challenges of large RSI perception, we construct a new benchmark called LRS-VQA (**L**arge **R**emote **S**ensing image Visual Question Answering). LRS-VQA features larger size images with lengths up to 27,328 pixels and 8 distinct question types. Our key contributions are as follows:

- We introduce a Region Focus Module (RFM) that distills the ability from LLM part of LVLM to efficiently identify text-related key vision tokens.
- Based on RFM and DIP, we propose an efficient coarse-to-fine inference strategy for large RSIs, which achieves a balance between resolution and efficiency through key tile selection and vision token pruning.
- We construct a new benchmark named LRS-VQA, featuring larger image sizes and more diverse question types than existing benchmarks, to reflect the challenges of large RSIs perception.
- The proposed method is architecture-agnostic and shows performance improvements and efficiency gains.

## 2. Related Work

### 2.1. LVLMs for High-Resolution Comprehension

Numerous LVLMs designed for high-resolution image or video perception have recently emerged. They can be primarily categorized into the following paradigms:

**Methods that carefully design image cropping and padding strategy.** Methods like [10, 19, 29, 39, 41, 45, 80, 90] traverse pre-defined partition grids and obtain the most suitable grid, to divide the high-resolution image into image tiles. These methods encounter limitations when processing large RSIs as in Fig. 1.

**Methods based on multiple visual branches.** These approach [34, 49, 63] employ additional high-resolution visual encoders like SAM [25] encoder or ConvNext [46] to process images with higher input resolution.

**Visual chain-of-thoughts methods.** These approaches utilize a search-and-focus strategy [59, 60, 66, 69] or external RAG pipeline [92] to identify text-related key image regions. However, such pipelines require multiple LVLM inferences, making them more complex and cumbersome.

### 2.2. Vision Token Reduction in LVLMs

Visual token pruning for transformers has been a classic research topic, it aims to accelerate computation by dynamically reducing less important tokens. It has been extensively studied in both natural language processing [24, 79] and computer vision domains [16, 26, 51, 56, 68].

Recently, a variety of token pruning methods have been proposed for LVLMs, which can be categorized into three main types: token pruning conducted in the vision encoder [5, 33, 58, 67, 77, 78], token pruning performed in the LLM component [8, 20, 21, 72, 81, 88], and collaborative approaches that integrate both [6, 30, 31, 36, 74, 82, 85]. Moreover, adaptive token segmentation methods [1, 7, 28] also offer a promising approach for token reduction.

Although vision-part pruning methods efficiently reduce vision tokens, when processing large images, it's highly time-consuming to traverse pre-defined grids and handle numerous image tiles with the vision encoder. Moreover, the absence of language guidance makes it hard to identify foreground regions in complex RSIs. The LLM-part pruning methods require feeding all vision tokens into the LLM. Therefore, long visual sequences from large images may exceed LLM length limits and incur high computational costs for attention score ranking.

Among the collaborative approaches, the methods most closely related to ours are LLaVA-Mini [85] and FlexAttention [31]. The former employs modality pre-fusion to compress visual information; however, its LLM-like pre-fusion module still struggles with handling extremely long vision sequences. The latter integrates text-guided high-resolution features within the LLM component. Nevertheless, the allowable image size for its high-resolution view remains limited, and it can only index high-resolution features once.

## 3. Preliminaries

In this section, we present the foundational concepts of text-image attention within the LLM component of



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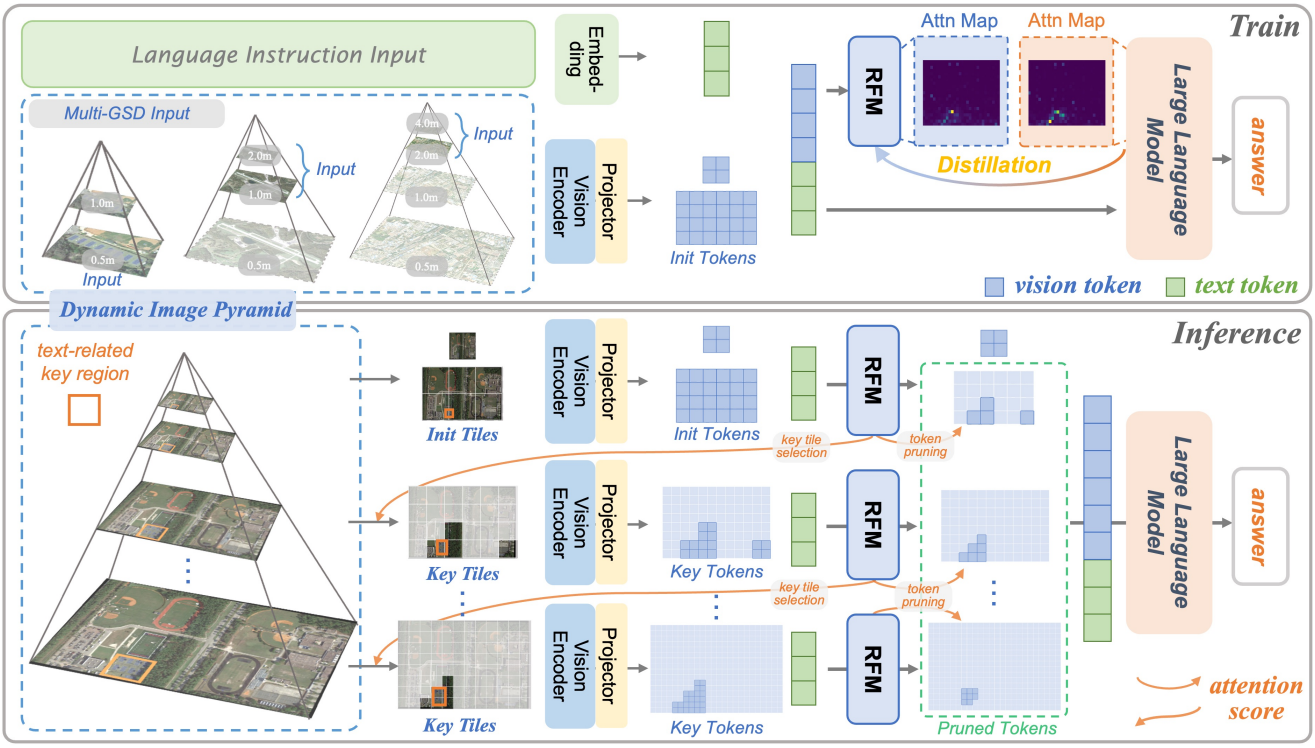



Figure 2. The pipeline of the proposed method. The entire process iterates in a coarse-to-fine manner, dynamically retrieving high-resolution features from the next DIP level (leftward orange arrow) or performing token pruning at the current level (rightward orange arrow) based on the output of the RFM module at each iteration. During training, the RFM distillation text-related attention from the LLM; during inference, RFM generates the attention scores for the input vision tokens. GSD means ground sample distance.

LVLMs. Mainstream modular LVLMs include three components [48]: a pre-trained vision encoder, a projector, and a pre-trained decoder-only LLM. Conventional high-resolution strategies [10, 19] process the original image $I_{img}$ into a thumbnail view $I_{\text{thumb}}$ and a set of image tiles $I_{\text{tiles}}$. Then the vision encoder and the projector transforms $I_{\text{thumb}}$ and $I_{\text{tiles}}$ into vision token embeddings $T_{vis} = [T_{vis}^{lr}, T_{vis}^{hr}]$, which contain low-resolution tokens $T_{vis}^{lr}$ from the $I_{\text{thumb}}$ and higher-resolution $T_{vis}^{hr}$ from the $I_{\text{tiles}}$. $T_{vis}$ are concatenated with text token embeddings $T_{txt}$ from the tokenized text instruction, forming a multimodal token sequence $T = [T_{vis}, T_{txt}]$. The LVLM processes $T$ via transformer-based decoder layers with causal self-attention, where the attention score $A$ can be represented as:

$$A = \text{Attention}(Q, K) = \text{Softmax}\left(\frac{QK^{\top}}{\sqrt{d}}\right), \qquad (1)$$

where $A \in \mathbb{R}^{n \times n}$ is the self-attention map, $Q, K \in \mathbb{R}^{n \times d}$ are the query and key matrices derived from the input embeddings through linear transformations, $n$ is the input sequence length, and $d$ is the feature dimension. As $A$ determines the weight each token assigns to the former tokens in the sequence, by analyzing the attention from text tokens to vision tokens, we can identify text-related important vision tokens. Some studies [23, 81] observe that in the deep layers of LLM part in LVLMs, attention is predominantly allocated to text-related vision tokens compared to other vision tokens. This insight inspired us to design a coarse-to-fine mechanism to focus on key image regions.

## 4. Method and Benchmark

### 4.1. Method

To achieve efficient understanding of ultra-high-resolution images, our core idea is **coarse-to-fine focusing**: we first gain a low-resolution overview, then iteratively zoom into text-related regions for details, as shown in Fig. 2. Implementing this necessitates a multi-resolution representation. Therefore, we first introduce the Dynamic Image Pyramid (DIP) in Sec. 4.1.1. Then we need to identify where to focus next. To solve this, we propose the Region Focus Module (RFM) to provide attention distribution for input vision tokens (Sec. 4.1.2). Finally, these components are integrated into an iterative refinement process (Sec. 4.1.3).

#### 4.1.1. The Construction of Dynamic Image Pyramid

As established, our coarse-to-fine strategy requires a multi-resolution representation of the image $I_{img}$. We realize this through the DIP, a structure designed to provide image tiles



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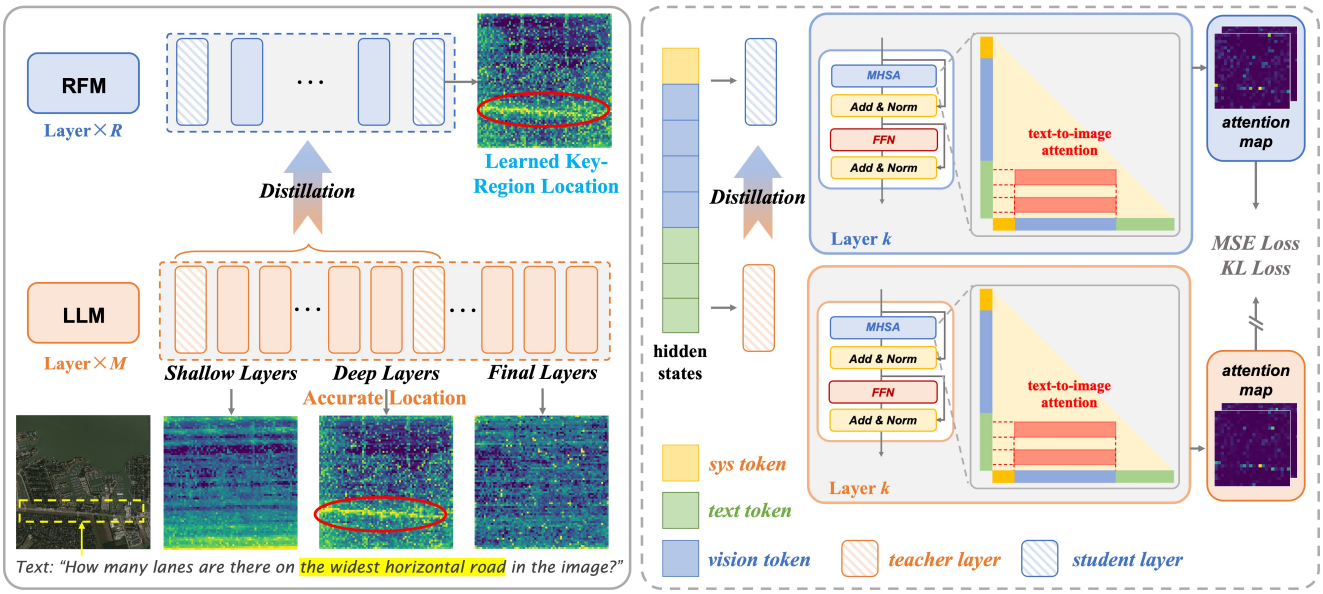



Figure 3. The proposed RFM and attention distillation strategy. The left part indicates our core idea: distill accurate text-related key region localization ability from the LLM part of the LVLM. The right part shows the distillation details. We only select specific layer pairs for distillation to avoid hidden state discontinuities. "sys token" represents the tokens from the system prompt.

at various ground sample distances (GSDs).

First, we iteratively downsample $I_{img}$ by a factor of 2, generating a series of images $\{I^1_{init}, I^2_{init}, \ldots, I^P_{init}\}$ until the shorter side reaches a pre-defined minimum length. Let a basic image tile size be $B \times B$ (e.g., $336 \times 336$ for CLIP-L14), for the $p$-th scaled image $I^p_{init}$, we calculate the number of image tiles along the height and width as follows:

$$N^p_h = \lceil H_p/B \rceil, \quad N^p_w = \lceil W_p/B \rceil, \tag{2}$$

where $H_p, W_p$ are the height and width of $I^p_{init}$, respectively. For proper tiling, we compute a scale factor $r_p$ as:

$$r_p = \min\left(\frac{N^p_h \times B}{H_p}, \frac{N^p_w \times B}{W_p}\right). \tag{3}$$

The image $I^p_{init}$ is then resized to $I^p$ using $r_p$, followed by padding to preserve the original aspect ratio. After resizing, $I^p$ is partitioned into non-overlap tiles $I^p_{\text{tiles}}$. Tiles from all scaled images are combined in reverse order, and integrated with the thumbnail view $I_{\text{thumb}}$ to form the final pyramid: $I_{DIP} = \{I_{\text{thumb}}, I^1_{\text{tiles}}, I^2_{\text{tiles}}, \ldots, I^{\text{P}}_{\text{tiles}}\}$, which consists of $P+1$ layers with progressively increasing resolutions. During training, we select the thumbnail $I_{\text{thumb}}$ and $I^1_{\text{tiles}}$ as visual inputs for computational efficiency. During inference, all pyramid layers become accessible to enable multi-scale inference, as in the left part of Fig. 2.

#### 4.1.2. Attention Distillation for Key Region Focus

With the DIP established, the central challenge becomes generating a reliable guidance signal to determine where to focus next. While a sliding-window similarity map from VLMs like CLIP is a straightforward baseline, we posit that the attention maps from the LVLM's LLM part provide a superior localization signal. This is inspired by prior work [23, 72, 81] where LLM attention tends to converge on key visual tokens. To support this, we quantitatively validate that LLM attention significantly outperforms the CLIP/RemoteCLIP baselines by offering more accurate, text-related localization signals in Tab. 7.

To harness this high-quality guidance efficiently, we introduce the Region Focus Module (RFM), a lightweight module trained via attention distillation to mimic the text-guided localization capability of the LLM teacher, as shown in Fig. 3. After distillation, the trained RFM can guide the selection of key image tiles and prune irrelevant vision tokens before the main LLM computation.

Specifically, assuming the LLM has $M$ layers $l_1, l_2, \ldots, l_M$, the proposed RFM adopts the same architecture with $R$ layers $r_1, r_2, \ldots, r_R$, where $R<M$. Each RFM layer $r_i$ is initialized from a selected LLM layer $l_{m_i}$, where $\{m_1, m_2, \ldots, m_R\}$ forms a sparse subset of LLM layer indices. For layer-wise distillation, we select $K$ pairs ($K<R$) from RFM and LLM as student-teacher pairs, represented as $\{(r_k, l_{m_k}) \mid k = 1, 2, \ldots, K\}$. This strategy avoids rigidly aligning each $r_i$ with $l_{m_i}$, which would neglect the non-teacher layers in the LLM, resulting in inconsistency between adjacent RFM layers.

As depicted in the right part of Fig. 3, the concatenated multimodal tokens $T = [T_{vis}, T_{txt}]$ are fed into both the RFM and the LLM during training. For the $k$-th student-teacher layer pair $(r_k, l_{m_k})$, we extract the attention scores between the last text token and all vision tokens from the



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multi-head self attention (MHSA). For the $h$-th head, the self-attention is represented as $A^{k,h}_{\text{stu}}, A^{k,h}_{\text{tea}} \in \mathbb{R}^{j \times n_v}$, where $j$ is the number of dialogue turns for multi-turn instruction and $n_v$ is the length of vision tokens. We apply the Kullback-Leibler (KL) divergence loss for distillation:

$$\mathcal{L}^h_{\text{kl}} = \sum_{k=1}^{K} \left[ D_{\text{KL}} \left( A^{k,h}_{\text{tea}} \parallel A^{k,h}_{\text{stu}} \right) + \lambda_{\text{hr}} D_{\text{KL}} \left( A^{k,h,\text{hr}}_{\text{tea}} \parallel A^{k,h,\text{hr}}_{\text{stu}} \right) \right], \tag{4}$$

where $A^{k,h,\text{hr}}_{\text{tea}}, A^{k,h,\text{hr}}_{\text{stu}}$ denote attention corresponding to $T^{hr}_{vis}$, and $\lambda_{\text{hr}}$ is the weighting factor. Additionally, we enforce stronger alignment constraints to the attention of $T^{hr}_{vis}$ by using an additional Mean Squared Error (MSE) loss:

$$\mathcal{L}^h_{\text{mse}} = \sum_{k=1}^{K} \text{MSE} \left( A^{k,h,\text{hr}}_{\text{stu}}, A^{k,h,\text{hr}}_{\text{tea}} \right). \tag{5}$$

The total distillation loss is computed as:

$$\mathcal{L}_{\text{distill}} = \frac{1}{KH} \sum_{h=1}^{H} \left( \lambda_{\text{mse}} \mathcal{L}^h_{\text{mse}} + \lambda_{\text{kl}} \mathcal{L}^h_{\text{kl}} \right), \tag{6}$$

where $K$ is the number of selected student-teacher layer pairs, $H$ is the number of attention heads, and $\lambda_{\text{mse}}, \lambda_{\text{kl}}$ are hyperparameters. For the flash-attention [13], we employ a specialized value matrix to extract attention maps like [88] to ensure compatibility.

#### 4.1.3. Text-Guided Token Pruning with Pyramid

Having introduced the DIP and the RFM, this section details how they are integrated into a cohesive, iterative process. This process achieves our coarse-to-fine strategy by selectively retrieving high-resolution tiles and pruning irrelevant tokens, as illustrated in Fig. 2.

The specific inference procedure, detailed in Alg. 1, is as follows. For simplicity, the tokens from the system prompt are omitted. We initialize the vision input using $I_{\text{thumb}}$ and $I^p_{\text{tiles}}$ to generate the initial $T_{vis} = [T^{lr}_{vis}, T^{p,hr}_{vis}]$, where $p = 1$, then the concatenated tokens $T$ are fed into the RFM. From the last layer of the RFM, we compute the average attention scores from all heads in the MHSA, and extract the attention scores $A^{K,hr}_{\text{stu}}$ of $T^{p,hr}_{\text{vis}}$. By selecting the top-$\lambda$ proportion from $A^{K,hr}_{\text{stu}}$, we identify the indices of key tokens and map their coordinates to image tile-level coordinates, to select key image tiles $I^{p+1}_{\text{key}}$ from $I^{p+1}_{\text{tiles}}$. Based on the number of $I^{p+1}_{\text{key}}$, we determine whether to prune the $T^{p,hr}_{\text{vis}}$ directly or replace it with vision tokens $T^{p+1,hr}_{\text{vis}}$ from higher-resolution tiles. If $T^{p+1,hr}_{\text{vis}}$ is obtained, the tile selection process is repeated recursively until the last layer of DIP is reached. This approach enables LVLM to focus only on processing a few high-resolution image tiles in a coarse-to-fine manner, thereby reducing computational complexity while preserving critical text-related image details.

### 4.2. The Construction of LRS-VQA Benchmark

Considering the limitations of existing benchmarks for evaluating LVLMs' perception of RSIs in terms of question diversity and image size, we propose a new benchmark for more comprehensive evaluation.

**Algorithm 1** Coarse-to-Fine Token Pruning with DIP

**Require:** $\{I^1_{\text{tiles}}, I^2_{\text{tiles}}, \ldots, I^{\text{P}}_{\text{tiles}}\}$ in $I_{\text{DIP}}$, $K$-layer RFM, initial multimodal tokens $T$, vision encoder and projector $\mathcal{V}(\cdot)$, token saving ratio $\alpha$, tile number threshold $N_{\max}$
**Ensure:** Retained tokens $T^{\text{retain}}_{\text{vis}}$

**Definition:** Top-$\alpha(X)$ selects elements in $X$ with values in the top $\alpha$ proportion.
**for** $p = 1$ **to** $P$ **do**
  Compute attention: $A^{K,hr}_{\text{stu}} \leftarrow \text{RFM}(T)[T^{p,hr}_{\text{vis}}]$
  Normalize:
  Identify key positions: $\mathcal{P}^{K,hr}_{\text{key}} \leftarrow \text{Top-}\alpha(A^{K,hr}_{\text{stu}})$
  **if** $p < P$ **then**
    Map coordinates and select key tiles: $I^{p+1}_{\text{key}} \leftarrow \{t \in I^{p+1}_{\text{tiles}} \mid \text{mapped from } \mathcal{P}^{K,hr}_{\text{key}}\}$
    **if** $|I^{p+1}_{\text{key}}| > N_{\max}$ **then**
      Prune tokens: $T^{\text{retain}}_{\text{vis}} \leftarrow \{t \in T^{p,hr}_{\text{vis}} \mid \mathcal{P}^{K,hr}_{\text{key}}\}$
      **break**
    **else**
      Encode selected tiles: $T^{p+1,hr}_{\text{vis}} \leftarrow \mathcal{V}(I^{p+1}_{\text{key}})$
      Update $T \leftarrow [T^{\text{lr}}_{\text{vis}}, T^{p+1,hr}_{\text{vis}}, T_{\text{txt}}]$
    **end if**
  **else**
    Prune tokens at final layer: $T^{\text{retain}}_{\text{vis}} \leftarrow \{t \in T^{p,hr}_{\text{vis}} \mid \mathcal{P}^{K,hr}_{\text{key}}\}$
  **end if**
**end for**
**return** $T^{\text{retain}}_{\text{vis}}$

#### 4.2.1. Annotation Pipeline

The annotation pipeline of LRS-VQA is illustrated in Fig. 4. We collect 3 publicly remote sensing datasets: FAIR1M-1.0 [62], GLH-Bridge [35], and STAR [37], and identify all unique targets based on the object detection labels to generate unique references from them (e.g., "the top-most ship" and "the largest dock"). Subsequently, we crop the region with the unique target from the original image, and feed it, along with a prompt containing the unique reference, into GPT-4V [54] to generate question-answer pairs about the target's color, shape, status, etc. Details of the overall construction process are provided in the Appendix A.

#### 4.2.2. Visual-Centric Quality Check and Refinement

To ensure the quality of the generated question-answer pairs, we employ powerful Qwen2-VL [65] for quality assessment. Qwen2-VL supports manual adjustment of the maximum pixel resolution, allowing us to evaluate whether increased resolution improves accuracy while keeping the LLM unchanged. Using this approach, we filter out ques-



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| Dataset | Image Size | Image Num | Ques. Cate. | Ques. Num | QS Format |
|---|---|---|---|---|---|
| MME-RealWorld-RS [91] | 689-11500 | 1298 | 3 | 3738 | Single-choice |
| LRS-VQA (Ours) | 1024-27328 | **1657** | **8** | **7333** | Open-end |

Table 1. Comparison of existing benchmarks for evaluating LVLMs' perception capabilities in large RSIs.

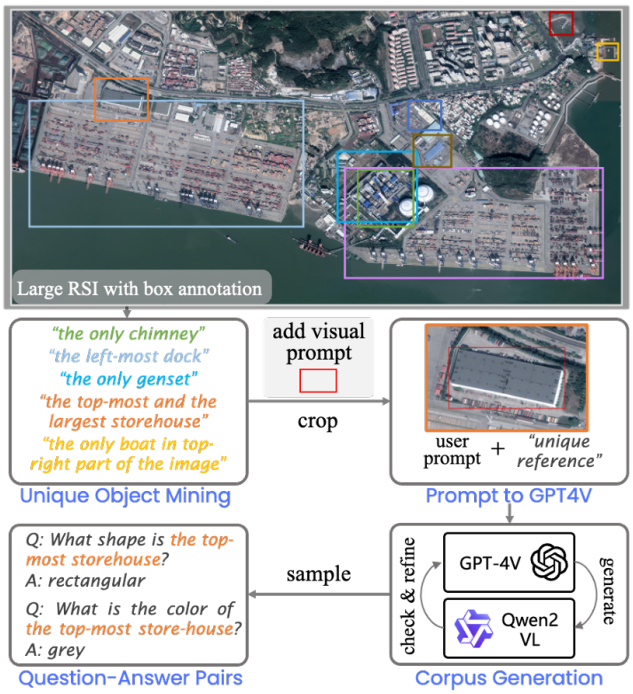



Figure 4. The construction pipeline of the proposed LRS-VQA dataset. The visual prompt (red box) is inspired by SoM [76].

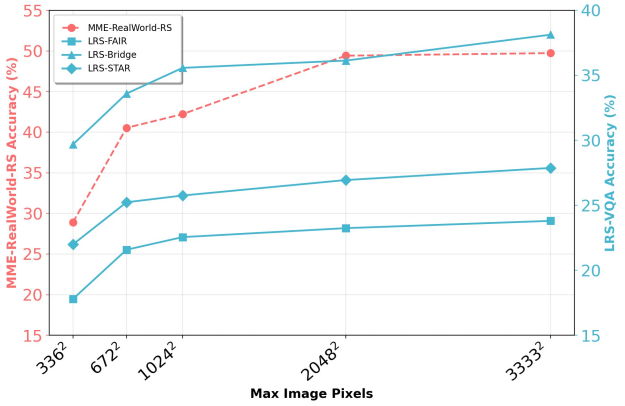



Figure 5. The accuracy trends of Qwen2-VL across varying input maximum pixels. This demonstrates that accuracy on both the manually annotated MME-RealWorld-RS and our proposed LRS-VQA exhibit a positive correlation with resolution improvement, proving the effectiveness of LRS-VQA in evaluating LVLM's high-resolution RSI perception capabilities.

tion types where accuracy can not improve with resolution. Through iterative filtering, manual review, and prompt refinement, we obtain the LRS-VQA dataset, which could effectively assess the large RSI perception capability of LVLMs, as shown in Fig. 5. LRS-VQA contains eight question-answer types: *count, color, category, shape, status, reasoning, rural/urban classification*, and *target background*. It features greater diversity in question types and larger image sizes than MME-Realworld-RS as in Tab. 1, highlighting the complexities of large RSI perception.

## 5. Experiments

### 5.1. Experimental Details

**Data.** Our pre-training (PT) phase is the same as LLaVA-1.5 [42] (using 558K data). For the SFT phase, we collect 484K samples, including 300K sampled from LLaVA-1.5-665K, 146K sampled from RSVQA-HR [47], and 38K template-based samples from three RS datasets [15, 35, 37] using their labels. Within this 38k data, some images share the same source with LRS-VQA, but there is no overlap between them. Details can be found in the Appendix B.1.

**Benchmarks and evaluation metrics:** i) MME-RealWorld-RS: the RS part of MME-Realworld [91], containing 1,298 RSIs with expert-annotated questions in three types: color, count, and position. We follow the official evaluation script but modify the prompt by removing "The best answer is:" to address Vicuna-1.5-based models' tendency to respond with only option A. ii) LRS-VQA: it consists of 3 parts: LRS-FAIR, LRS-Bridge, and LRS-STAR, containing 2,272, 1,062, and 3,999 QA pairs, respectively. For the short open-ended format, we adopt a structured evaluation metric following [22, 71], using WordNet [52], with a semantic similarity threshold of 0.8.

**Experimental settings.** We evaluate existing LVLMs using their official weights or APIs. For fair comparison, we also re-implement some methods on our 484k SFT dataset. Our approach is built upon the LLaVA-1.5 codebase, where we integrate the anyres strategy and Qwen2 language model. All experiments are conducted on 4 A100 80GB GPUs. For our method, the minimum length for DIP is 1,008 pixels. The $N_{\max}$ is set to 40 and 80 for Vicuna-1.5 and Qwen2, respectively. The vision token saving ratio $\alpha$ is 0.25, $\lambda_{\text{hr}}, \lambda_{\text{mse}}, \lambda_{\text{kl}}$ are set to 2.0, 1.0, 1.0. The RFM-LLM layer pairs are $[1, 5, 11, 14]$ with distillation applied to the first and last pairs. Additional experimental details are provided in Appendix B.2.



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| Leaderboard | Data | Vis. Encoder | LLM | Max Size | MME-RW-RS | LRS-FAIR | LRS-Bridge | LRS-STAR | Avg. Acc |
|---|---|---|---|---|---|---|---|---|---|
| Qwen2-VL [65] | - | QwenViT | Qwen2-7B | 3,333×3,333 | 49.73 | 23.80 | 38.12 | 27.87 | 34.88 |
| LLaVA-OV [29] | 4.8M | SigLip | Qwen2-7B | 2,304×2,304 | 53.53 | 20.61 | 35.11 | 26.08 | 33.83 |
| IXC-2.5 [84] | - | CLIP-L14 | InternLM2-7B | 4,096×4,096 | 36.12 | 25.25 | 38.41 | 27.30 | 31.77 |
| LLaVA-UHD-v2 [89] | 1.42M | CLIP-L14+JBU | Vicuna-1.5-7B | 672×1,008 | 40.77 | 22.82 | 32.57 | 26.08 | 30.56 |
| LLaVA-FlexAttn [31] | 1.22M | CLIP-L14 | Vicuna-1.5-7B | 1,008×1,008 | 37.75 | 19.57 | 29.99 | 22.76 | 27.52 |
| SEAL [69] | 0.95M | CLIP-L14 | Vicuna-1.5-7B | - | 30.55 | 21.29 | 34.75 | 21.29 | 26.97 |
| MGM-HD [34] | 3.0M | Mixed | Vicuna-1.5-7B | 1,536×1,536 | 31.51 | 17.90 | 35.92 | 20.13 | 26.36 |
| SliME [90] | 2.0M | CLIP-L14 | Vicuna-1.5-7B | 672×1,008 | 28.54 | 17.11 | 32.09 | 22.99 | 25.18 |
| LLaVA-1.5 [41] | 1.22M | CLIP-L14 | Vicuna-1.5-7B | 336×336 | 26.38 | 18.76 | 30.70 | 22.63 | 24.62 |
| RSUniVLM [44] | 1.2M | SigLIP | Qwen2-0.5B | 336×336 | 25.39 | 21.02 | 32.61 | 24.72 | 25.93 |
| Geochat [27] | 1.53M | CLIP-L14 | Vicuna-1.5-7B | 504×504 | 28.62 | 20.18 | 24.54 | 13.75 | 21.77 |
| GPT-4o [3] | - | - | - | - | 28.92 | 22.15 | 31.84 | 27.40 | 27.58 |
| GPT-4o-mini [3] | - | - | - | - | 6.69 | 18.67 | 31.99 | 25.85 | 20.80 |
| Claude-3.5-Sonnet [4] | - | - | - | - | 25.74 | 12.95 | 26.69 | 13.29 | 19.67 |
| **Comparison** | | | | | | | | | |
| *LLaVA-1.5** | 1.04M | CLIP-L14 | Vicuna-1.5-7B | 336×336 | 34.40 | 18.24 | 32.28 | 24.17 | 27.30 |
| *LLaVA-1.5-p4** | | CLIP-L14 | | 672×672 | 38.20 | 19.18 | 35.43 | 26.50 | 29.83 |
| *Geochat** | | CLIP-L14 | | 504×504 | 33.68 | 21.51 | 35.97 | 25.86 | 29.25 |
| *SLiME** | | CLIP-L14 | | 672×1,008 | 34.56 | 21.98 | 33.20 | 25.10 | 28.71 |
| *LLaVA-UHD-v2** | | CLIP-L14+JBU | | 672×1,008 | 38.55 | 20.77 | 36.57 | 25.79 | 30.42 |
| Ours (LLaVA-1.5) | | CLIP-L14 | | Dynamic | **39.04** | **22.97** | **36.89** | **27.48** | **31.59** |
| *LLaVA-Next-p4** | 1.04M | CLIP-L14 | Qwen2-7B | 672×672 | 39.12 | 21.06 | 36.27 | 25.80 | 30.56 |
| *LLaVA-Next-p9** | | | | 1,008×1,008 | 40.35 | 21.14 | 37.25 | 26.10 | 31.21 |
| *LLaVA-Next-p25** | | | | 1,680×1,680 | 39.65 | 20.99 | 36.38 | 26.18 | 30.80 |
| Ours (LLaVA-Next) | | | | Dynamic | **41.89** | **21.85** | **38.24** | **26.67** | **32.16** |

Table 2. Leaderboard and performance comparison on LRS-VQA and MME-RealWorld-RS. "MME-RW-RS" indicates the MME-RealWorld-RS. "Data" refers to the total instruction data used during PT and SFT. "$pX$" indicates the max number is $X$ for the pre-defined grids. "*method**" indicates we reproduce the SFT stage of existing methods using the 484k instruction data.

| Setting | Color | Count | Position | Acc | FPS |
|---|---|---|---|---|---|
| anyres-p25 | 41.56 | 31.05 | 46.14 | 39.65 | 0.188 |
| *w/ PruMerge++* [58] | 43.27 | 28.38 | 32.78 | 34.86 | 0.152 |
| *w/ VisionZip* [77] | 42.71 | 24.55 | 34.55 | 33.98 | 0.183 |
| *w/ FastV* [8] | 39.52 | 30.10 | 43.99 | 37.93 | 0.192 |
| *w/ PDrop* [72] | 41.00 | 30.54 | 47.77 | 39.85 | 0.184 |
| *w/ prune (CLIP)* | 37.80 | 27.65 | 44.90 | 36.86 | 0.171 |
| *w/ prune (RemoteCLIP)* | 39.40 | 29.83 | 47.45 | 38.77 | 0.148 |
| *w/ prune (Ours)* | 43.98 | 30.42 | 49.16 | 41.28 | 0.165 |
| DIP-3layer *w/ prune (Ours)* | 44.08 | 30.94 | 49.26 | **41.31** | **0.267** |

Table 3. Comparison with different token reduction methods based on LLaVA-Next-Qwen2. The "anyres-p25" setting means the maximum number of pre-defined grids is 25 (1680×1680 pixels). FPS is computed based on the inference time for continuous 800 images from MME-Realworld-RS.

## 5.2. Leaderboard and Comparison

**Leaderboard on LRS-VQA.** We evaluate a total of 11 open-source LVLMs, including methods for high-resolution [31, 42, 84, 90], chain-of-thought reasoning [69], multiple visual encoders [34], and RS LVLMs [27, 44]. We also evaluate 3 closed-source MLLMs. Results in Tab. 2 indicate that LRS-VQA is more challenging than MME-RealWorld-RS on average, reflecting the complexities of large RSIs.

| Setting | Vis. Tokens (Total) | Vis. Tokens (to LLM) | TFLOPs (B) |
|---|---|---|---|
| anyres-p144 | 83520 | 21312 | 243.37 |
| *w/ PruMerge++* [58] | 83520 | 5328 | 43.75 |
| *w/ VisionZip* [77] | 83520 | 9280 | 83.56 |
| *w/ FastV* [8] | 83520 | 21312 | 109.21 |
| *w/ PDrop* [72] | 83520 | 21312 | 101.62 |
| *w/ prune (Ours)* | 83520 | 5760 | 82.56 |
| DIP-4layer *w/ prune (Ours)* | **55296** | **2376** | **36.61** |

Table 4. Comparison of computational efficiency with different token reduction methods based on LLaVA-Next-Qwen2. We assume inferring a 4000×4000 pixel image (anyres-p144) and report the theoretical TFLOPs of vision tokens after the vision projector.

**Comparison with high-resolution methods.** As shown in Tab. 2, we select 5 different high resolution strategies, performing SFT on the same dataset for comparison. Our method performs better across all 4 datasets.

**Comparison with token reduction methods.** We reproduce 4 plug-and-play methods and establish 2 key baselines for comparison on the MME-RealWorld-RS. These baselines utilize sliding-window similarity maps generated



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| Method | Drop Ratio | Color | Count | Pos | Acc |
|---|---|---|---|---|---|
| anyres-p25 | 0% | 41.56 | 31.05 | 46.14 | 39.65 |
| *w/ prune (Ours)* | 25% | 42.47 | 31.08 | 47.41 | 40.40 |
| | 50% | 42.39 | **31.24** | 48.45 | 40.77 |
| | 75% | **43.98** | 30.42 | **49.16** | **41.28** |
| | 90% | 41.51 | 29.77 | 46.70 | 39.41 |

Table 5. Ablation study on different prune ratios with LLaVA-Next-Qwen2, when pruning vision tokens based on RFM results without DIP. This table demonstrates that pruning irrelevant tokens in high-resolution RSIs can improve performance.

| Max Size | Color | Count | Pos | Acc | FPS |
|---|---|---|---|---|---|
| 1,008×1,008 | 42.87 | 29.85 | 47.89 | 40.29 | **0.271** |
| 2,016×2,016 | 44.08 | 30.94 | 49.26 | 41.31 | 0.267 |
| 4,032×4,032 | **44.86** | 29.85 | 47.18 | 40.72 | 0.221 |
| 8,064×8,064 | 43.98 | 30.42 | 45.11 | 39.91 | 0.204 |
| Dynamic | 44.70 | **31.00** | **49.72** | **41.89** | 0.238 |

Table 6. Ablation study on fixing the number of DIP layers reached in inference with LLaVA-Next-Qwen2. Rows 1-4 correspond to fixing the number of DIP layers being 2-5, respectively.

by CLIP-L14 [55] and RemoteCLIP-L14 [40], respectively. More details are in the Appendix B.2. As shown in Tab. 3, our method achieves the best performance under the fixed anyres-p25 resolution. It also secures the highest accuracy and FPS in the 3-layer DIP setting (up to 2,016×2,016 pixels). For Tab. 4, we report the average number of tiles selected by our dynamic method for images of similar sizes. These results suggest that existing methods often incur high computational costs when processing numerous text-irrelevant image tiles in large images.

### 5.3. Ablation Results

**Different ratios for RFM-based pruning.** As shown in Tab. 5, a key finding is that in large RSIs, the complex background and the small foreground regions enable high pruning rates to deliver performance benefits.

**Fixed DIP layers for inference.** As shown in Tab. 6, we fix the number of DIP layers reached during inference. If an image's DIP is shallower than the specified number, the last layer is used. Results show that forcing traversal to higher resolutions can degrade performance since not all questions require image details. Therefore, our strategy dynamically selects the termination DIP layer based on the input text, balancing accuracy and efficiency.

**Attention Localization validation.** We select LRS-VQA (using boxes of cropped regions) and two referring-based datasets, DIOR-RSVG [83] and RRSIS-D [43] for localization accuracy evaluation, using a **recall metric**: a prediction is considered successful if the vision tokens retained

| Pruning Guidance | LRS-VQA | DIOR-RSVG | RRSIS-D |
|---|---|---|---|
| Teacher LLM Attn. | 56.14 | 74.81 | 73.88 |
| CLIP Sim. | 20.14 | 29.10 | 27.53 |
| RemoteCLIP Sim. | 30.41 | 43.12 | 41.81 |
| RFM Attn. (Ours) | **47.99** | **64.76** | **61.17** |

Table 7. Localization recall (%) of different pruning guidance.

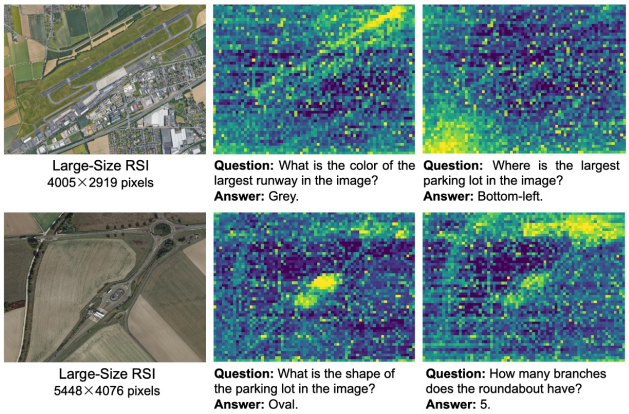



Figure 6. Text-related visual attention localization in the last layer if trained RFM under LLaVA-Next-Qwen2.

after pruning cover more than 50% of the ground-truth region. The results in Tab. 7 demonstrate that at a 25% token retention ratio, both the teacher LLM's attention and our student RFM achieve higher localization accuracy than the CLIP-L14 and RemoteCLIP-L14 baselines (also based on sliding-window similarity maps). This confirms that LLM attention provides a reliable localization signal and our distillation is effective.

**Visualization results.** In Fig. 6, we visualize the attention maps generated by the trained RFM for vision tokens from initial input image tiles. The results demonstrate that text references to targets effectively guide the RFM's attention output, supporting our coarse-to-fine inference.

## 6. Conclusion

This paper presents a text-guided token pruning method tailored for efficiently processing large remote sensing images (RSIs). To address the limitations of existing high-resolution approaches when handling large RSIs, our method integrates the Region Focus Module (RFM) and Dynamic Image Pyramid (DIP) to focus on text-relevant key vision tokens while reducing computational overhead. Furthermore, we introduce a new benchmark, LRS-VQA, with 7,333 QA pairs covering 8 question types, and features larger image sizes and a diverse range of question types.

**Future work.** Vision-language understanding of large RSIs remains a challenging task. Investigating approaches such as long-context transfer from LLMs to LVLMs, or chain-of-thought reasoning, is also meaningful.



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## Acknowledgment

The authors would like to thank the anonymous reviewers for their valuable comments. This work was supported by the National Natural Science Foundation of China under Grants 42371321, 424B2006, and the Natural Science Foundation of Shanghai under Grant No. 25ZR1402268.

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